

Competitive Exam Practice 1: Computer Architecture,
Organization, and Performance
GATE (Computer Science) Pattern

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type.

- Q1** [*GATE CS 2014 Set 1 Q55, verified — GATEOverflow V3 p.91*] [**NAT, 1 mark**] Consider two processors P_1 and P_2 executing the same instruction set. Under identical conditions, for the same input, a program running on P_2 takes 25% less time but incurs 20% more CPI (clock cycles per instruction) than the program running on P_1 . If the clock frequency of P_1 is 1 GHz, the clock frequency of P_2 (in GHz) is _____.
- Q2** [*GATE CS 2024 Set 1 Q45, verified — GATEOverflow V3 p.91*] [**NAT, 2 marks**] The baseline execution time of a program on a 2 GHz single-core machine is 100 ns. The code corresponding to 90% of the execution time can be fully parallelized. The overhead of using an additional core is 10 ns when running on a multicore system. Assume all cores run their share of the parallelized code for an equal amount of time. The number of cores that minimizes the execution time of the program is _____.
- Q3** [*GATE CS 2014 Set 1 Q43, verified — GATEOverflow V3 p.91*] [**NAT, 2 marks**] Consider a 6-stage instruction pipeline with all stages perfectly balanced and no cycle-time overhead from pipelining. When an application executes on this 6-stage pipeline, the speedup achieved with respect to non-pipelined execution, if 25% of the instructions incur 2 pipeline stall cycles each, is _____.
- Q4** [*GATE CS 2026 Set 2 Q47, verified — GATEOverflow V3 p.91*] [**NAT, 2 marks**] A non-pipelined instruction execution unit operating at 1.6 GHz takes an average of 5 clock cycles to complete the execution of an instruction. To improve performance, the system was pipelined with the goal of one instruction per clock cycle, but could operate only at 1.2 GHz due to pipeline overheads. While executing a program on the pipelined design, 30% of instructions encountered a stall of 2 cycles due to pipeline hazards. The speedup obtained by the pipelined design over the non-pipelined one for this program is _____ (rounded off to two decimal places).
- Q5** [*Original, GATE-pattern*] [**MCQ, 1 mark**] Two machines, M_1 and M_2 , both run at 1 GHz. For a fixed program, M_1 reports 800 MIPS and M_2 reports 400 MIPS. M_1 has a CPI of 1.25; M_2 has a CPI of 2.5. Which statement is correct?

(A) M_1 is twice as fast as M_2 on this program

- (B) M_2 is twice as fast as M_1 on this program
- (C) They run the program in equal time, since f/CPI determines time and f is equal
- (D) MIPS alone cannot be compared because the instruction sets may differ

Q6 [*Original, GATE-pattern*] [MCQ, 2 marks] A program spends 30% of its execution time on a part that is being upgraded. Which of the following is **TRUE** about the maximum possible overall speedup?

- (A) It is unbounded if the upgraded part becomes infinitely fast
- (B) It approaches $1/(1 - 0.30) \approx 1.43\times$ as the upgrade becomes infinitely fast
- (C) It is at most $0.30\times$ the upgrade factor
- (D) It is exactly $3.33\times$ regardless of the upgrade factor